Evolutionary Computation Assignment

The attached file **TSPDATA.txt** shows the coordinates of ۱۲۷ different cities on a map. Assuming that a direct (Euclidean) path exists between every pair of cities, write an evolutionary algorithm to solve the Traveling Salesman Problem (TSP) for these cities with the minimum total distance. The salesman should start from one city, visit all cities exactly once, and end at the final point with the shortest possible total distance.

At the end, report the best route found and its total distance. Also report the number of algorithm iterations required to reach this solution.

For this task, prepare a report and repeat the experiment using:

- Two crossover methods: **Edge Recombination** and **PMX** (**Partially Matched Crossover**)
- Two mutation methods: **Swap** and **Insert**
- Two different selection methods: **Fitness-Proportionate** (**Roulette Wheel**) and **Linear Rank Selection**

(There will be ۸ different combinations in total.)

Compare the results with each other. Which methods produce better results? The comparisons must be based on performance after a fixed number of iterations, for example, ۱۰۰۰ iterations. That is, after *n* iterations, which method achieves a better result?

The remaining parameters are up to you to choose.

You may use any programming language.